

Personal FM News Channel Using Raspberry Pi and RSS



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Ever wished for a personal FM channel that reads out to you the news items of your choice over a radio or podcast? I always wished for my very own personal FM channel-cum-station at home that I could schedule with my favourite RSS feeds to be broadcast so as to reduce my overall screen exposure daily. I found that Raspberry Pi was a perfect platform to do that. Here is how you can set up your own FM news channel using Raspberry Pi.

Basic components

The project is simple and can be implemented in a short time. The basic components required are:

RSS feed source. RSS stands for 'really simple syndication.' It is an old standard using which various websites allow their users to stay posted about new content, without the need to really visit the website. Typical examples of the websites that provide RSS feeds are news websites and blog sites.

RSS feeds are typically HTTP URLs. You need Internet connectivity to access them and fetch the content.

Internet connectivity. As mentioned above, you need Internet connection on your Raspberry Pi to fetch the RSS feeds and channel these to the FM transmitter after converting them into speech format.

Raspberry Pi. Raspberry Pi is at the heart of this setup but it is not really mandatory; you can also use your home desktop or laptop. The sheer simplicity, size and power consumption needs of Raspberry Pi prompted me to use it for this project. It is assumed that you have used Raspberry Pi earlier and are familiar with its setup and configuration options. For

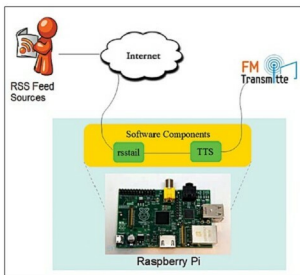


Fig. 1: Overview of the Raspberry Pi-based FM news channel

this project, we shall leverage its audio-out port.

Software components. Software components play an important part in this setup as we need means to fetch the news articles from RSS source and convert them into voice that can then be streamed into the FM transmitter and finally to your own radio.

Software components include:

rsstail. rsstail is a generic Linux utility program that allows you to crawl any RSS feed URL and fetch its RSS articles.

TTS. TTS stands for text-to-speech. There are numerous libraries available for various platforms across languages, 'festival' and 'espeak' being the most popular for Linux users. Refer software section for pointers on installing these packages. There are also various articles/blogs on the Internet on how to specifically get the TTS test programs running on Raspberry Pi. TTS programs take any text input (lines or strings of characters) and generate an audio output stream that can be channeled to any audio speaker device con-

nected to your Raspberry Pi audio jack.

FM transmitter. FM transmitter is the final piece of the puzzle. There are various kinds and makes of FM transmitters available in the market for fairly reasonable prices. I opted for an FM transmitter that takes input from any standard audio-out jack and transmits at your configured frequency.

Software

This section covers the various commands and utility scripts that we shall put together to get the project going. Before getting started, it is recommended that you have knowledge of UNIX basics like root login, telnet/ssh, bash scripting, IP address and man command.

Step 1: Start and login to Raspberry Pi. Power up your Raspberry Pi and login to it using an SSH client (refer 'Remote Access to Raspberry Pi' article published in EFY June issue).

Default login on Raspberry Pi is 'pi.' Change it to 'root' using the below-mentioned command as you will need root privileges for further installations:

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$ sudo su
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Step 2: Install the required tools. The tools required are rsstail, espeak and alsa-utils. Install these as follows:

rsstail. As mentioned earlier, rsstail is needed to crawl over any RSS source URL and extract the various elements of RSS articles (published time, title, description, etc).

To install rsstail, type the following command as a root user on Raspberry Pi: